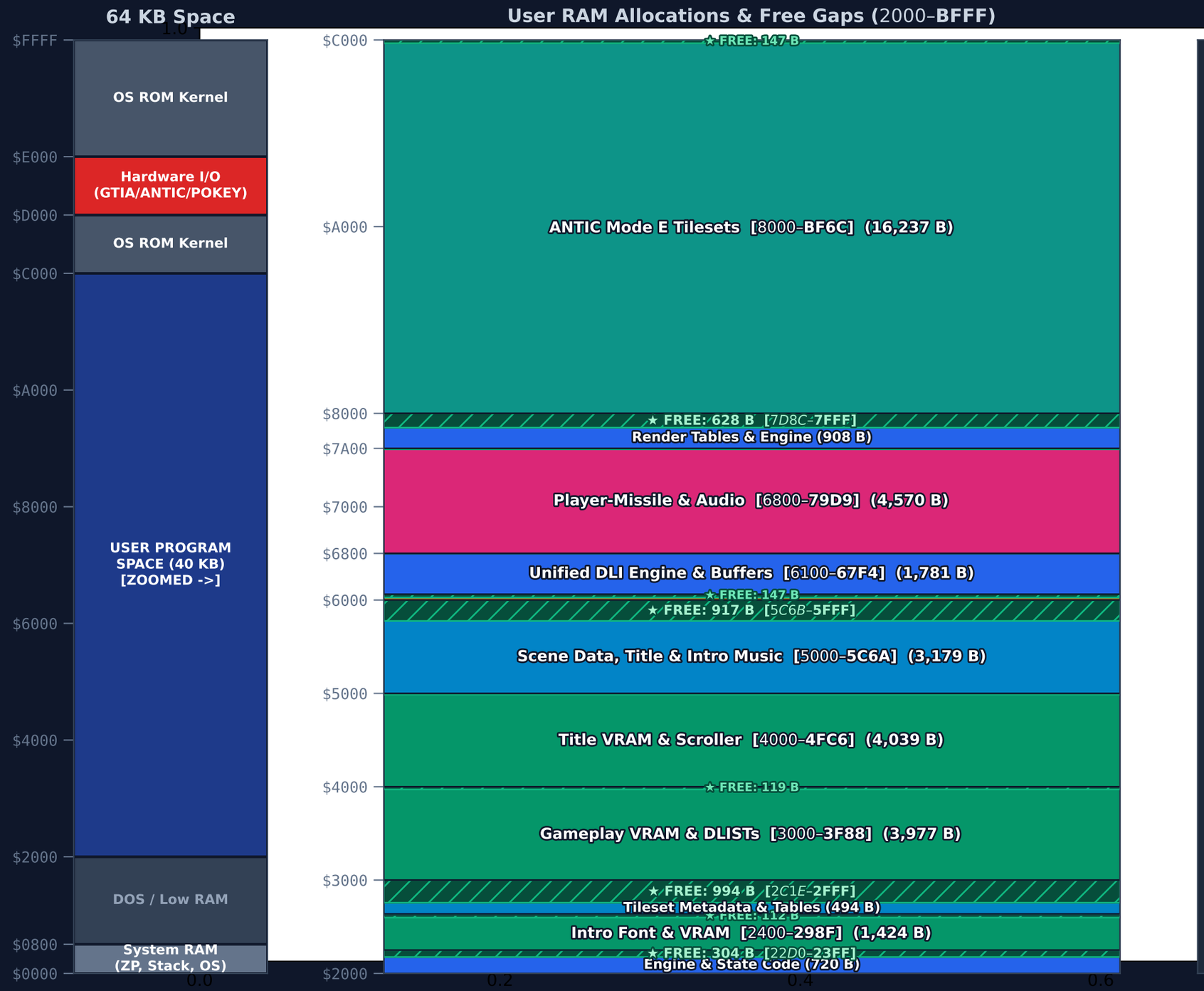
ATARI 8-BIT MEMORY MAP — CALL OF THE WINTERMON

File: call_of_the_wintermon.xex | Target: Atari 800XL / 65XE (MOS 6502) | RAM: 64 KB | User RAM: 2000–BFFF (40 KB)



MEMORY UTILIZATION SUMMARY

User RAM Space (2000–BFFF):	**40,960 Bytes (40.0 KB)**
Allocated Memory:	**37,468 Bytes (91.5%)**
Total Free Headroom:	**3,492 Bytes (8.5%)**
Largest Free Block:	**994 Bytes (2C1E–2FFF)**
Free Zero Page RAM:	**63 Bytes (C1–FF)**

CATEGORY LEGEND

CODE	Game Engine ($2000), DLI Engine ($6100), Render ($7A00)
GRAPHICS	Playfield VRAM (3000)&*Title/Scroller*(4000)
BITMAPS	ANTIC Mode E Tileset Sheets ($8000)
DLIST	Display Lists (Game $3F00, Title $6000, Intro $2A00)
PMG / AUDIO	Player-Missiles (6800)&*AudioPlayer*(7000)
METADATA	Scene SoA Tables, Title Code & Intro Music ($5000)
FREE RAM	Available unallocated memory gaps
HARDWARE / ROM	GTIA/ANTIC/POKEY (*D*000)&*OSROM*(C000+)

TOP FREE MEMORY REGIONS (HEADROOM)

1. 2*C1E*–**2FFF (994 Bytes / 0.97 KB)**
   Low RAM gap before Gameplay VRAM ($3000) — vacated by scenes data
2. 5*C6B*–**5FFF (917 Bytes / 0.90 KB)**
   Free expansion headroom in 5000*blockbeforeTitleDLIST*(6000)
3. 7*D8C*–**7FFF (628 Bytes / 0.61 KB)**
   Free headroom in 7000*blockbeforeANTICModeETilesets*(8000)
4. 22*D0*–**23FF (304 Bytes / 0.30 KB)**
   Free headroom in 2000*blockbeforeIntroFont*(2400) — vacated by title code
5. 606*D*–**60FF (147 Bytes / 0.14 KB)**
   Free space after Title DLIST before DLI double buffers ($6100)

ZERO PAGE BREAKDOWN (00–FF)

00–**7F (128 B)**	Reserved for Atari OS Kernel
80–**8F (16 B)**	Engine temporaries & state machine
90–**99 (10 B)**	POKEY Audio Player base pointers
9*A*–**B7 (30 B)**	Scene background renderer pointers
B8–**C0 (9 B)**	DLI engine registers & pattern pointers
C1–**FF (63 B)**	★ 100% FREE for Entities, AI, Physics